PCEOT402
ELECTRICAL MACHINES

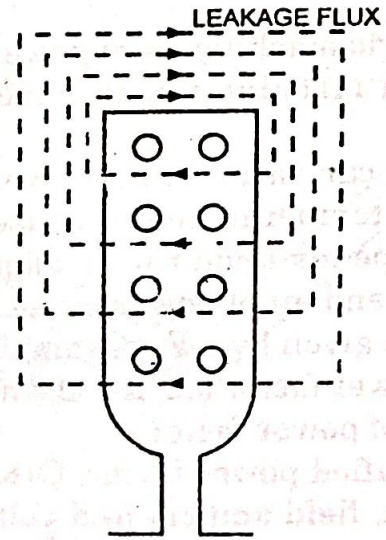
Module 3

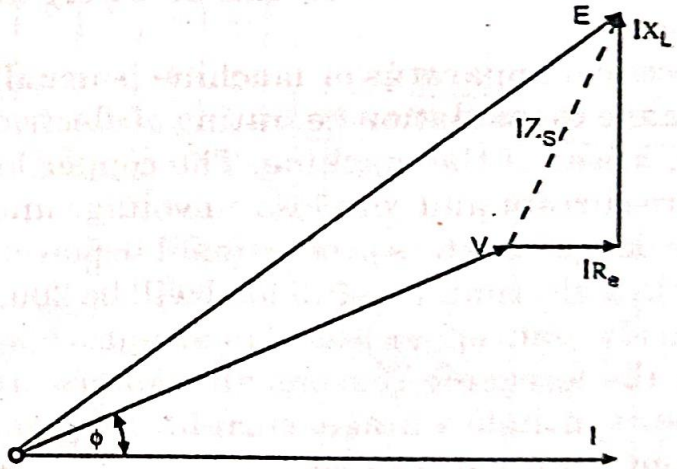
Synchronous Machines

Three phase Synchronous Machine

- Synchronous generator on no-load – open circuit characteristics –
- synchronous generator on load – armature reaction – effect of armature reaction
- Equivalent circuit - phasor diagram – voltage regulation –
- predetermination of voltage regulation by emf and mmf method
- Parallel operation - synchronous generator on infinite bus-bar –
- conditions – methods of synchronisation – dark lamp method – bright lamp method

Leakage reactance

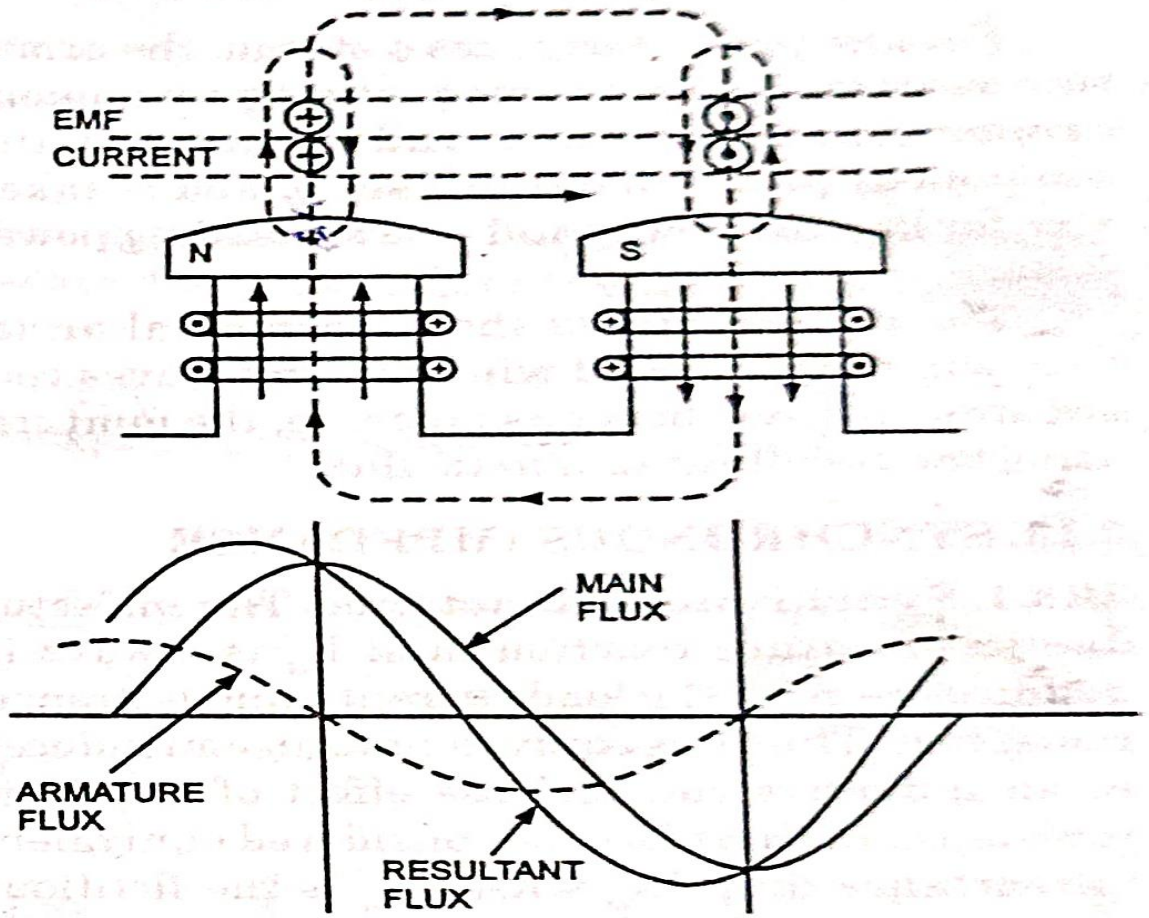




Armature Reaction

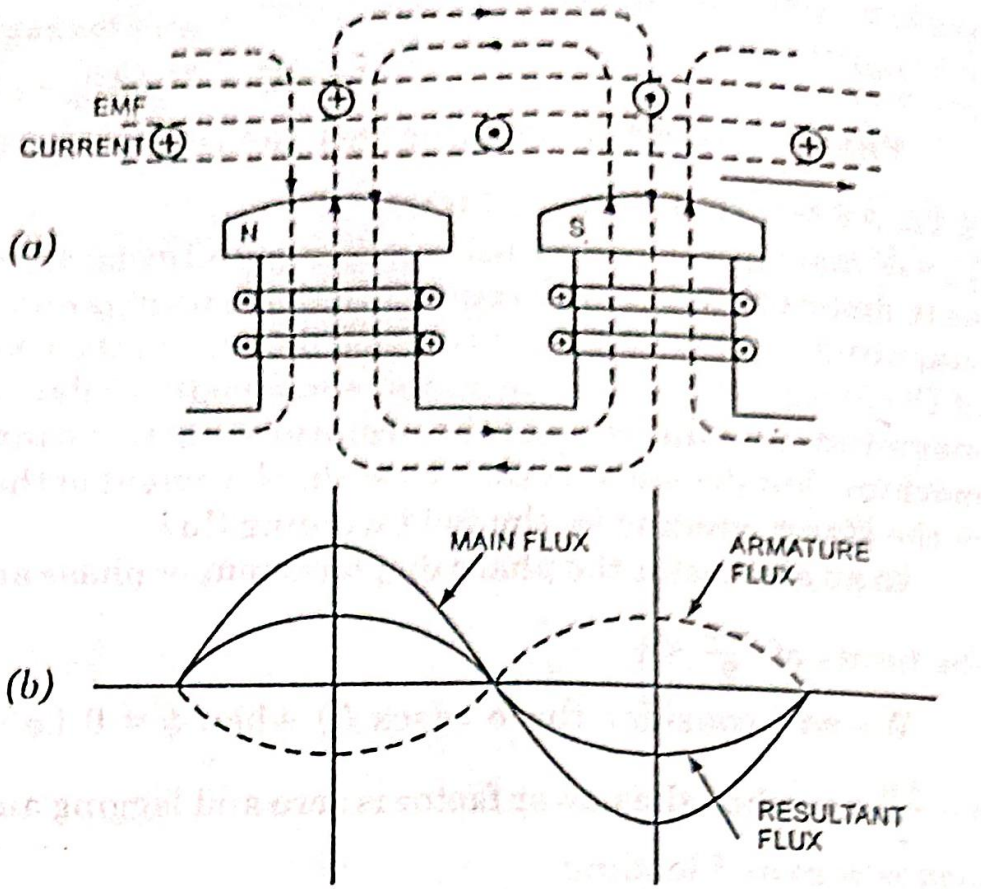
- Effect of armature flux on main flux
- In alternator it depend on type of load
- Effects are
    - Magnetising
    - Demagnetising
    - Cross magnetising

i) At upf load



Hence armature reaction at unity power factor has got distorting effect.

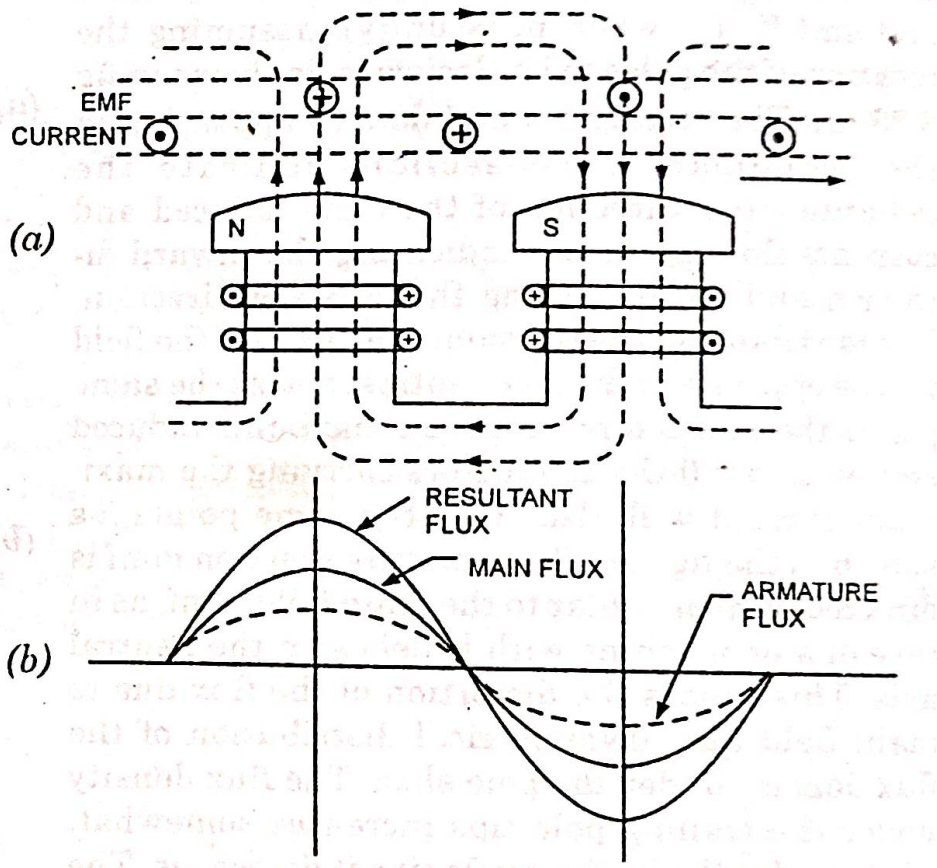
ii) Zero power factor lagging



Armature Reaction At Zero Power Factor (Lagging)

demagnetising effect.

iii) Zero power factor leading



Armature Reaction At Zero Power Factor (Leading)

Synchronous Impedance

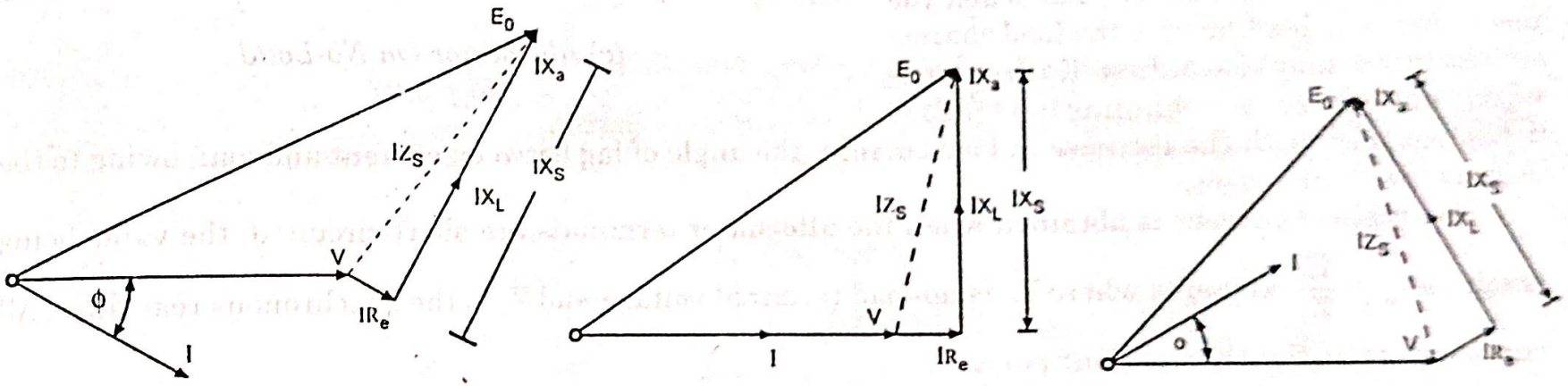
- Synchronous Reactance

$$X_s = X_L + X_a$$

- Synchronous Impedance

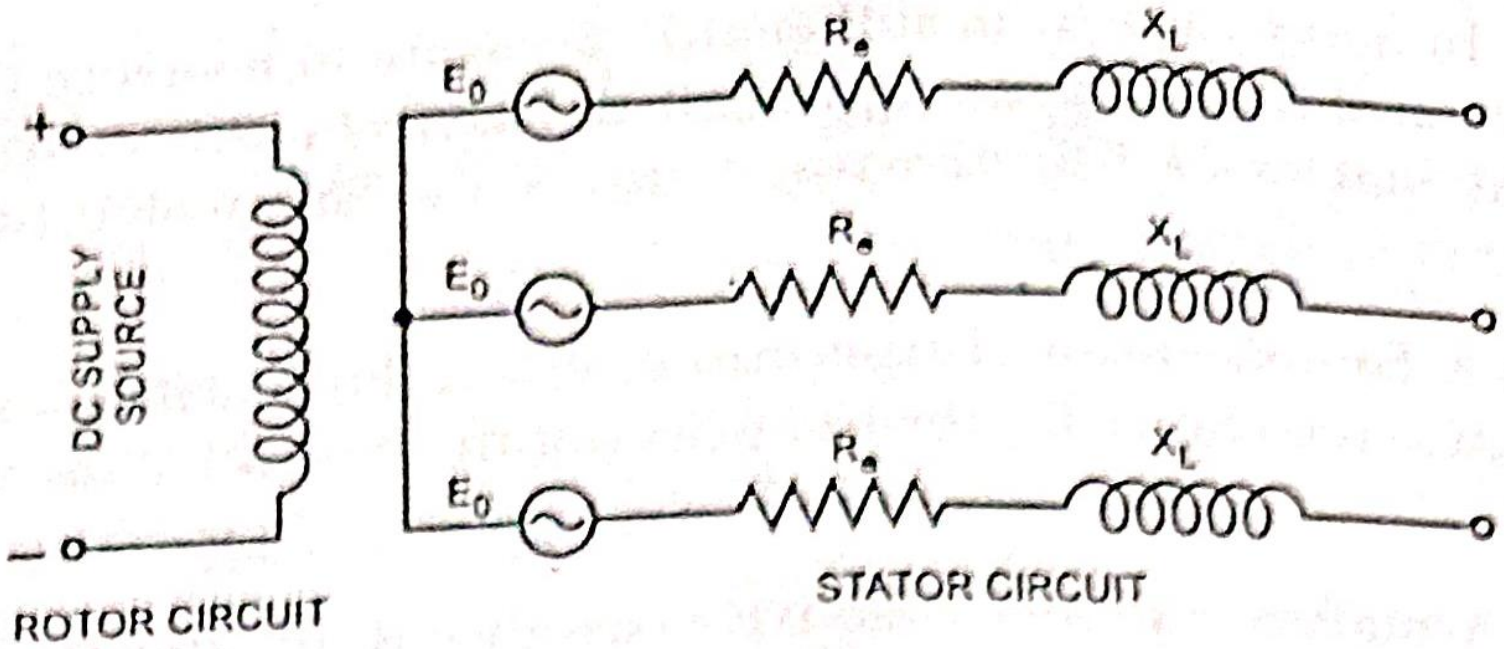
$$Z_s = R_a + jX_s$$

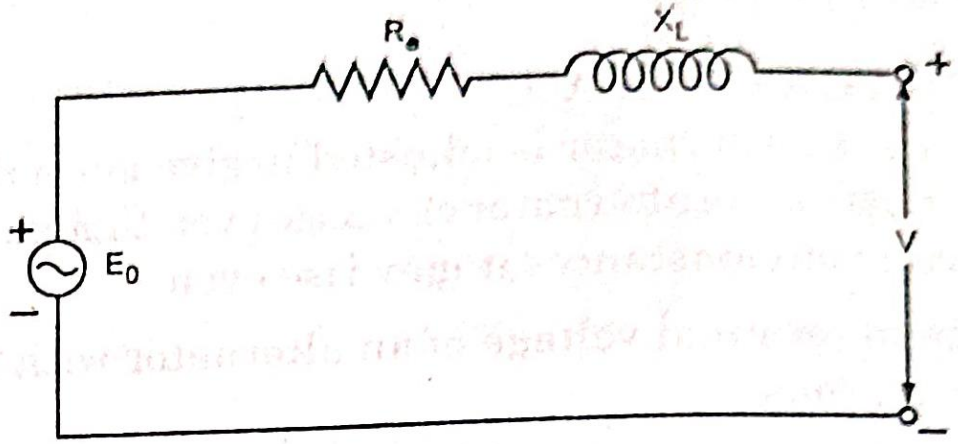
Alternator on load Phasor Diagram



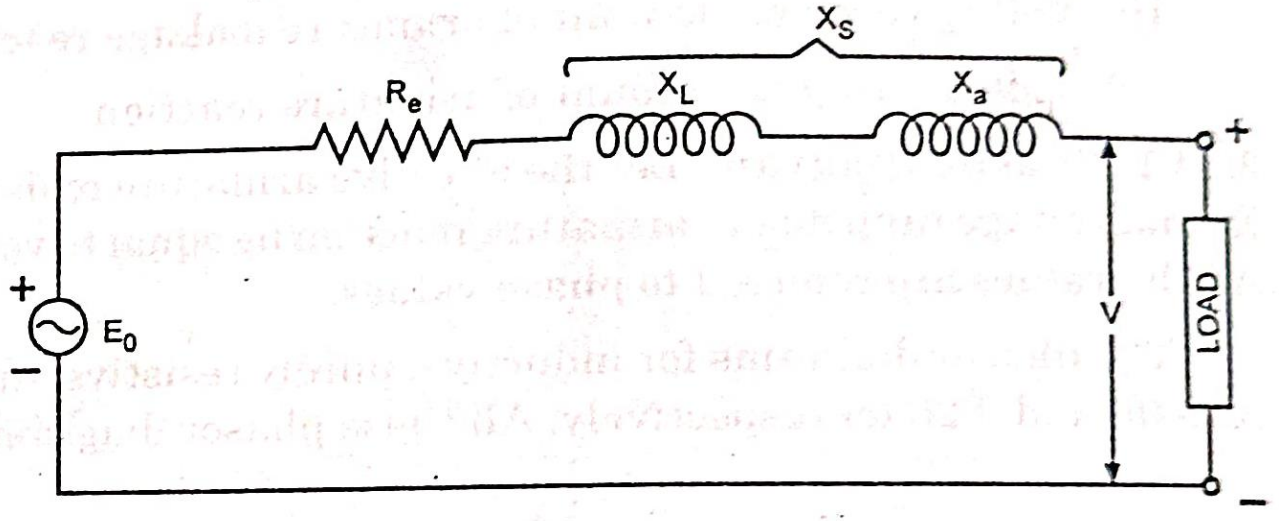
(a) At Lagging PF *(b) At Unity PF* *(c) At Leading PF*

Equivalent Circuit



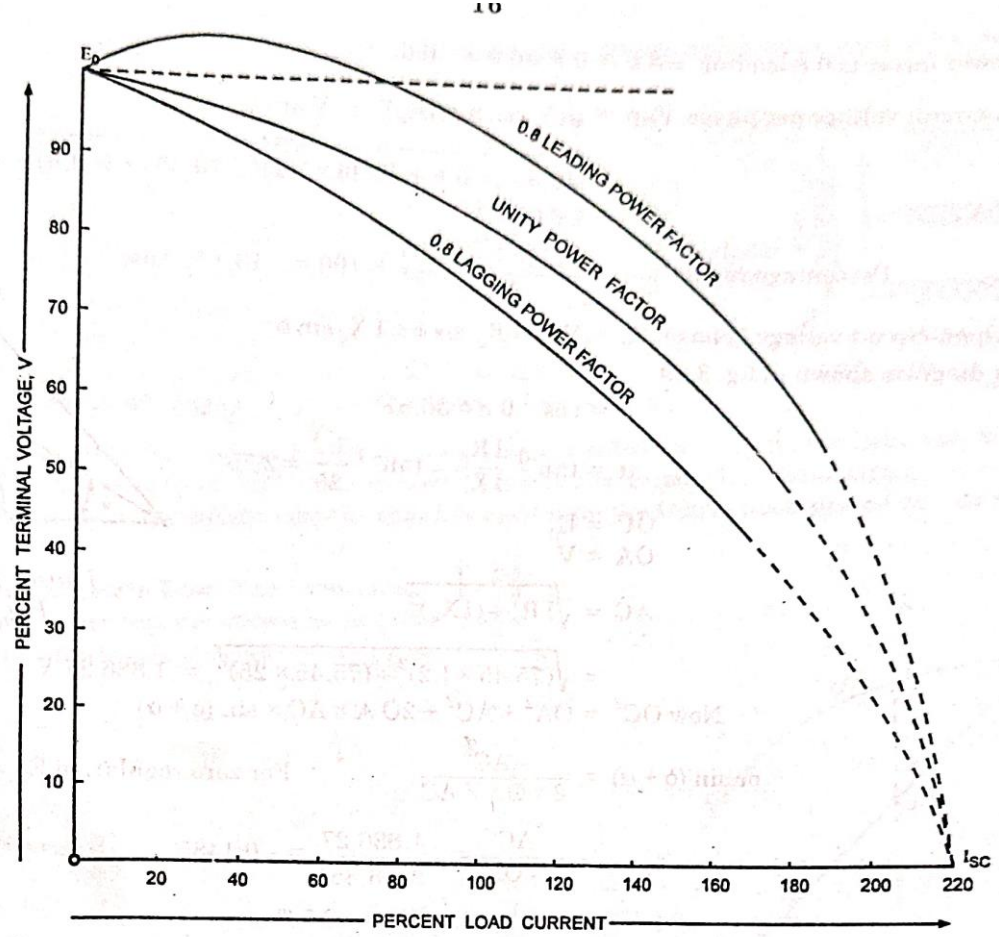


(b) Equivalent Circuit Diagram



(c) Alternator On No-Load

Load Characteristics



Voltage Regulation

i.e. voltage regulation $= \dfrac{E_0 - V}{V}$ per unit

$= \dfrac{E_0 - V}{V} \times 100$ in percentage

EMF Method (Synchronous Impedance Method)

nd short-circuit characteristic.

From short-circuit and open-circuit tests, the voltage egulation can be determined by any one of the two methods iven below.

1. Synchronous Impedance Method or EMF lethod. After determining R_a and X_s from open-circuit and ort-circuit tests, the phasor diagram for any load of any wer factor can be drawn, as shown in fig 3.35.

Let load current be I lagging behind the terminal phase ltage V by ϕ

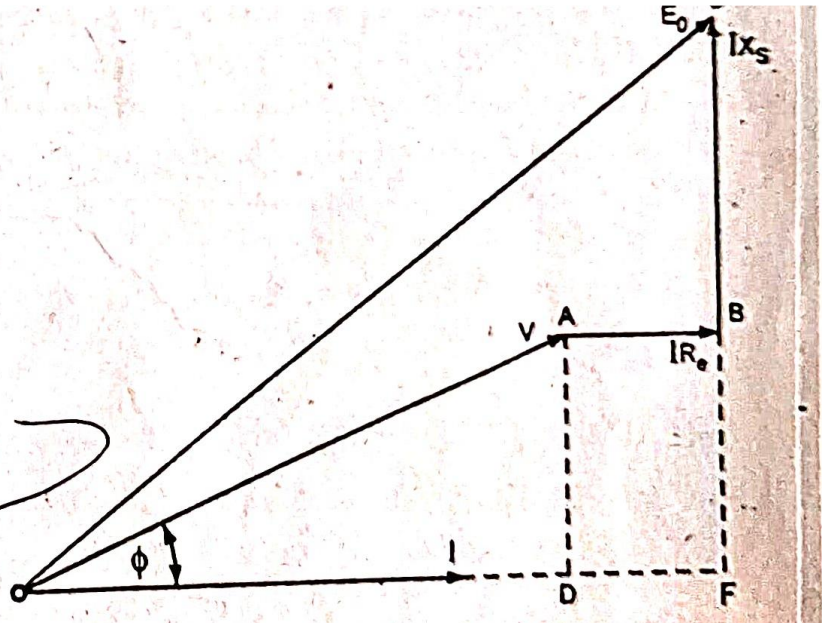


Fig. 3.35

No load terminal voltage per phase, E_0 = OC

Fulll-load terminal voltage, V = OA

Armature resistance drop, $I R_a$ = AB

Synchronous reactance drop $I X_s$ = BC

Now $OC^2 = OF^2 + FC^2 = (OD + DF)^2 + (FB + BC)^2$

or $E_0^2 = (V \cos \phi + I R_a)^2 + (V \sin \phi + I X_s)^2$

[Since BF = AD = V sin ϕ]

$$\text{or } E_0 = \sqrt{(V\cos\phi + IR_e)^2 + (V\sin\phi + IX_s)^2}$$

$$\text{And \% regulation} = \frac{E_0 - V}{V} \times 100$$

Note 1. For leading pf taking ϕ as negative and for unity pf taking $\phi = 0$ voltage regulation determined at leading pf and unity pf respectively.

2. Of various methods for determination of regulation, this one yields results most likely to be The regulation obtained by this method is always higher than actual value and, therefore, this is ca *pessimistic* method. The results are, however, more likely to be 'on safe side'. However, this m theoretically accurate for non-salient pole machines with distributed field windings when saturati considered. The reason of error is that in this method synchronous impedance or reactance is ass remain constant while actually it is not. The value of synchronous impedance (or reactance) varies saturation. At low saturation its value is larger because the effect of armature reaction is greater th at high saturation. Now under short-circuit conditions, saturation is very low and, therefore, the synchronous impedance measured is higher than that in actual working conditions

MMF (Ampere Turn Method)

synchronous impedance measured is higher than that in actual working conditions.

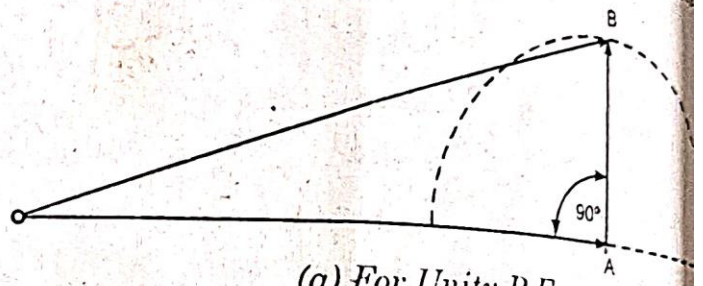
2. *Ampere-Turn (or MMF) Method.* In this method, the datas determined from open-circuit te
utilized.

From the above two tests the open-circuit and short-circuit characteristics are drawn, as shown in fig 3.32.

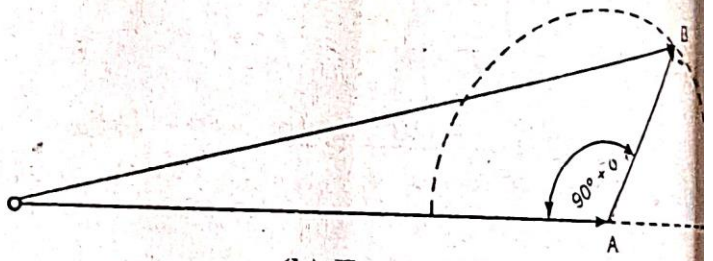
From the above characteristic field current I_{f1} is determined to give rated voltage V on no load, neglecting armature resistance drop, and field current I_{f2} is determined to cause short-circuit current, equal to full load current, on short circuit.

On short circuit, the field excitation I_{f2} balances the impedance drop in addition to armature reaction on full load. But since R_e is usually very small and X_L is also small for low voltage on short circuit, so impedance drop can be neglected. Hence pf on short circuit is almost zero lagging and the field amp-turns are used entirely to overcome the armature reaction. Therefore, I_{f2} gives demagnetising amp-turns at full load.

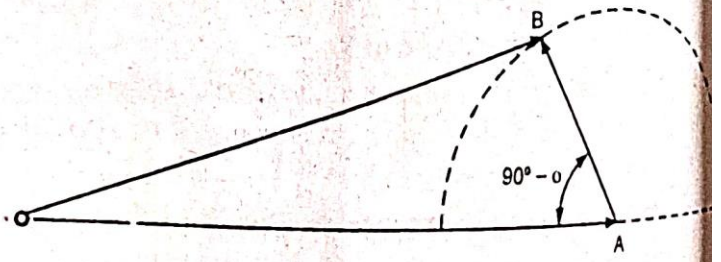
Now let the alternator supply full load current at a pf of cos ϕ. Draw OA representing I_{f1} to give full load rated voltage, V [or more exactly V + $I R_e \cos \phi$] then draw AB at an angle ($90^\circ \pm \phi$) representing I_{f2} to give full load current on short circuit; + ve sign for lagging pf and –ve sign for leading pf. Now find field current I_f measuring OB, which will give open-circuit EMF E_0, which can be determined from OCC.



(a) For Unity P F



(b) For Lagging P F



(c) For Leading P F

Parallel Operation of Alternators

- Efficiency
- Reliability or Continuity of Service
- Maintenance and Repair
- Physical Size

Requirements for Parallel Operation

- ✓ Alternators must have same output voltage rating
- ✓ Speeds of the machines should be such as to generate same frequency
- ✓ Alternators should be of same type so as to generate voltages of same waveform
- ✓ Prime movers of the alternators should have same speed-load characteristics
- ✓ Alternators should have reactances in their armature

Conditions for Synchronising

* **Synchronising** - The process of connecting an alternator in parallel with another alternator or with the common bus bar.

* For satisfactory operation,

- The terminal voltage of incoming machine must be exactly equal to that of running machines or common bus bar
- The speed of incoming machine must be such that its frequency *f*

equals to bus bar frequency

- Phase of incoming machine voltage must be the same as that of the bus bar voltage relative to the load.
- For three phase alternator, the phase sequence of incoming machine must be same as that of bus bar.

Methods of Synchronisation

- Synchronising of three phase alternators
  - Three Dark Lamp Method
  - Bright lamp Method (Two Bright and One Dark Lamp Method)
  - Synchroscope

Synchronization of 1-Phase Alternators

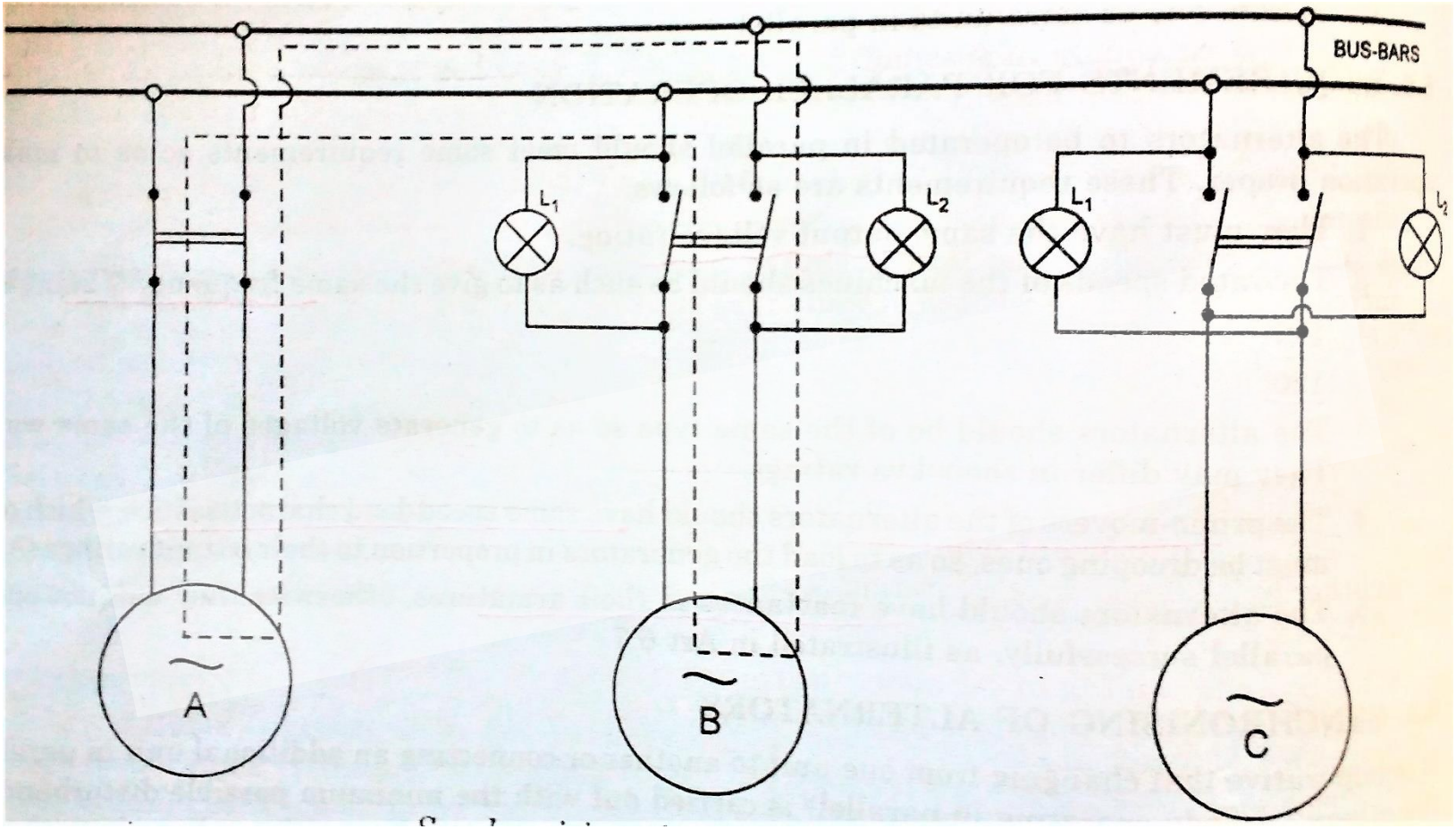


Figure 8 : Synchronising of single phase alternators

Dark Lamp Method

- Equality of terminal voltages can be determined by connecting voltmeters to the incoming and running alternators.
- Equality of frequency and phasing can be determined with the help of synchronising lamps.

Let alternator B is to be paralleled with alternator A,

- Prime mover of Alternator B is started and brought up close to the rated speed.
- Alternator B is excited and its voltage is raised by increasing the excitation.
- If frequencies of alternators A & B are exactly same and their terminal voltages w.r.t local series circuit → no resultant voltage will act across lamps L_1 and L_2 → Lamps will remain dark.
- If frequencies of A & B are not equal → flickering of lamps. Frequency of flickering = Difference of frequencies of two alternators.
- $\Rightarrow$ greater the difference in frequencies, greater will be the frequency of flick

Dark Lamp Method

- Synchronising is done at the middle of dark period.

  "At the time of synchronising of alternators, the speed of the incoming machine is adjusted until the lamps go in and out very slowly, the terminal voltage of the incoming alternator is made equal to bus bar voltage by adjusting the excitation of the incoming alternator and then switch of incoming alternator is closed in the middle of the dark period."

- But it is not so easy to judge the middle of dark period.

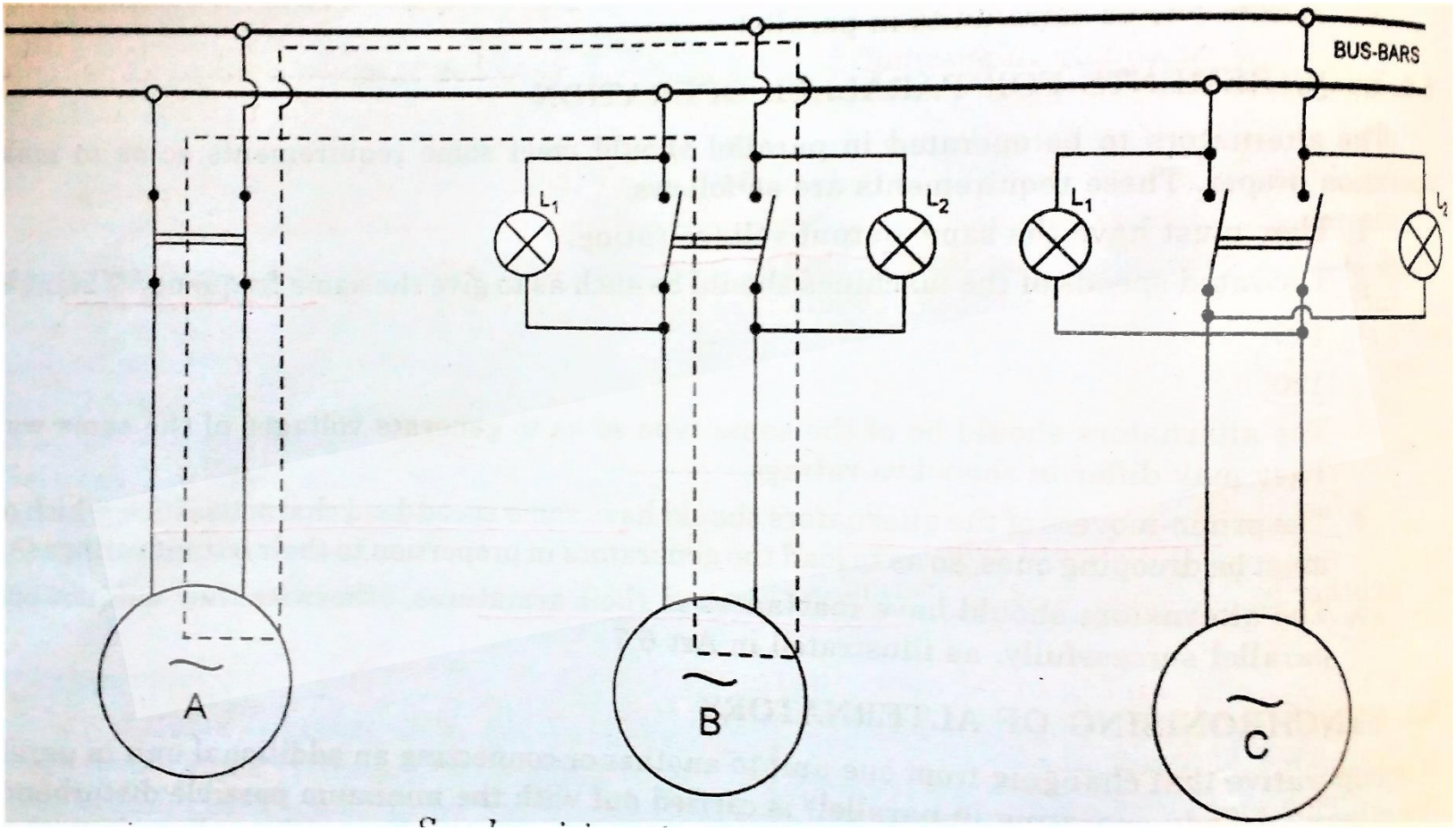


Figure 9 : Synchronising of single phase alternators

Bright Lamp Method

- Easy to judge the middle of bright period
- In **Bright lamp method**, lamps are cross-connected.
- Maximum voltage across lamps will occur when alternator C is in phase opposition with alternator A.
- Magnitude of maximum voltage that can exist across a lamp is double that of the voltage of one machine.
- After doing the adjustments, the synchronising switch is closed at the middle of bright period.

Synchronization of 3-Phase Alternators

"Phasing out the alternator" → The process of checking the phase sequence and getting it correct

1. Three Dark Lamp Method
2. Two Bright and One Dark Lamp Method
3. Using Synchroscope

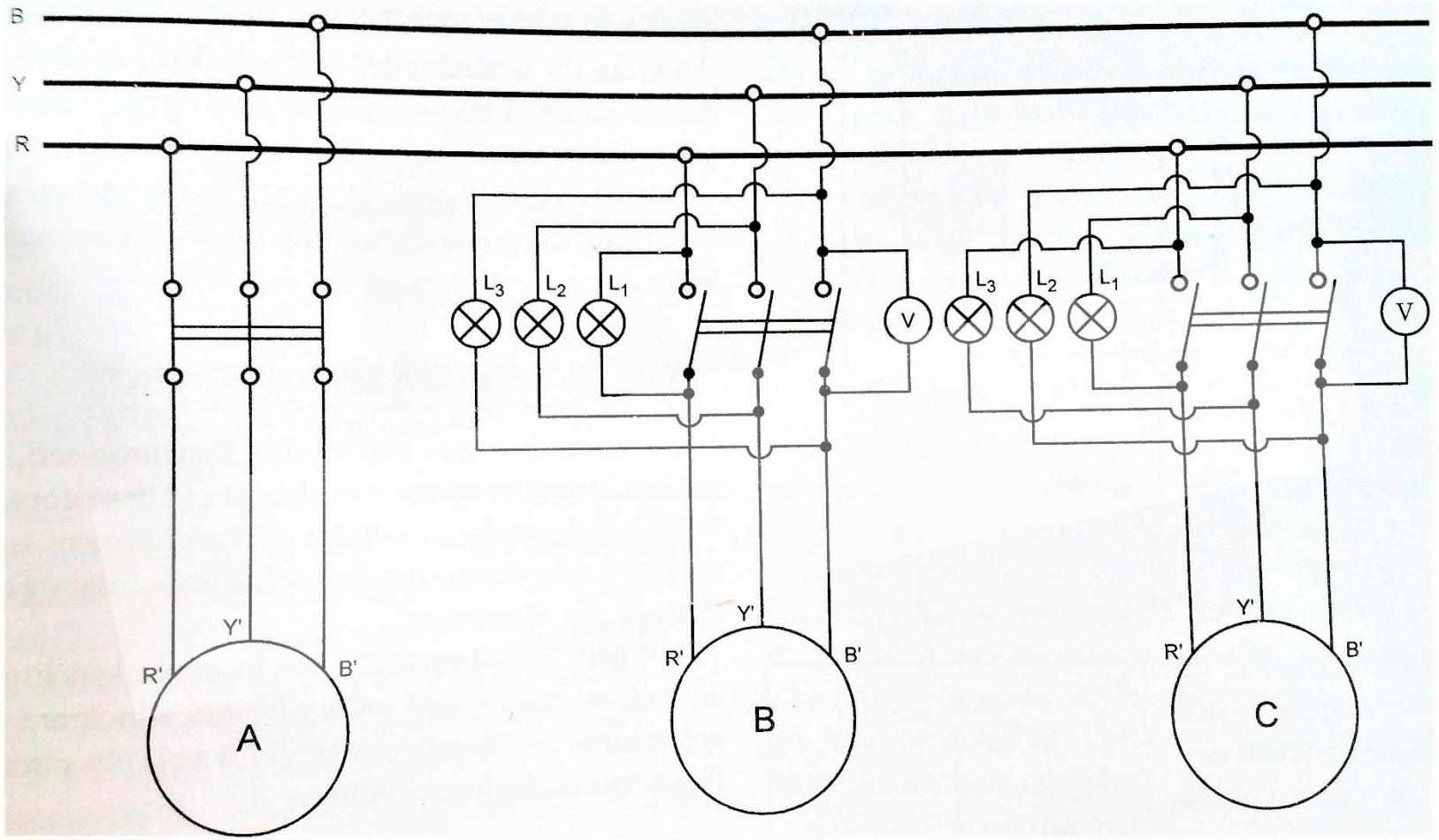


Figure 10 : Synchronising of 3-Phase Alternators

Three Dark Lamp Method

- Prime mover of alternator B is started and brought up close to the rated speed.
- Then alternator B is excited and its voltage is raised by increasing the excitation.
- If the incoming alternator B is properly connected, all the three lamps should become bright and dark together.
- If they bright and dim in sequence $\Rightarrow$ incoming alternator B is not properly connected with the bus-bars and the phase sequence of incoming alternator B must be reversed relative to the system.(Interchange any two leads on either the alternator side or the line side of the switch)
- The speed of incoming machine is further adjusted until the lamps flicker at a very rate and voltage is made equal to the bus bar voltage by adjusting the excitation.
- The synchronising switch is closed at the instant all the three lamps

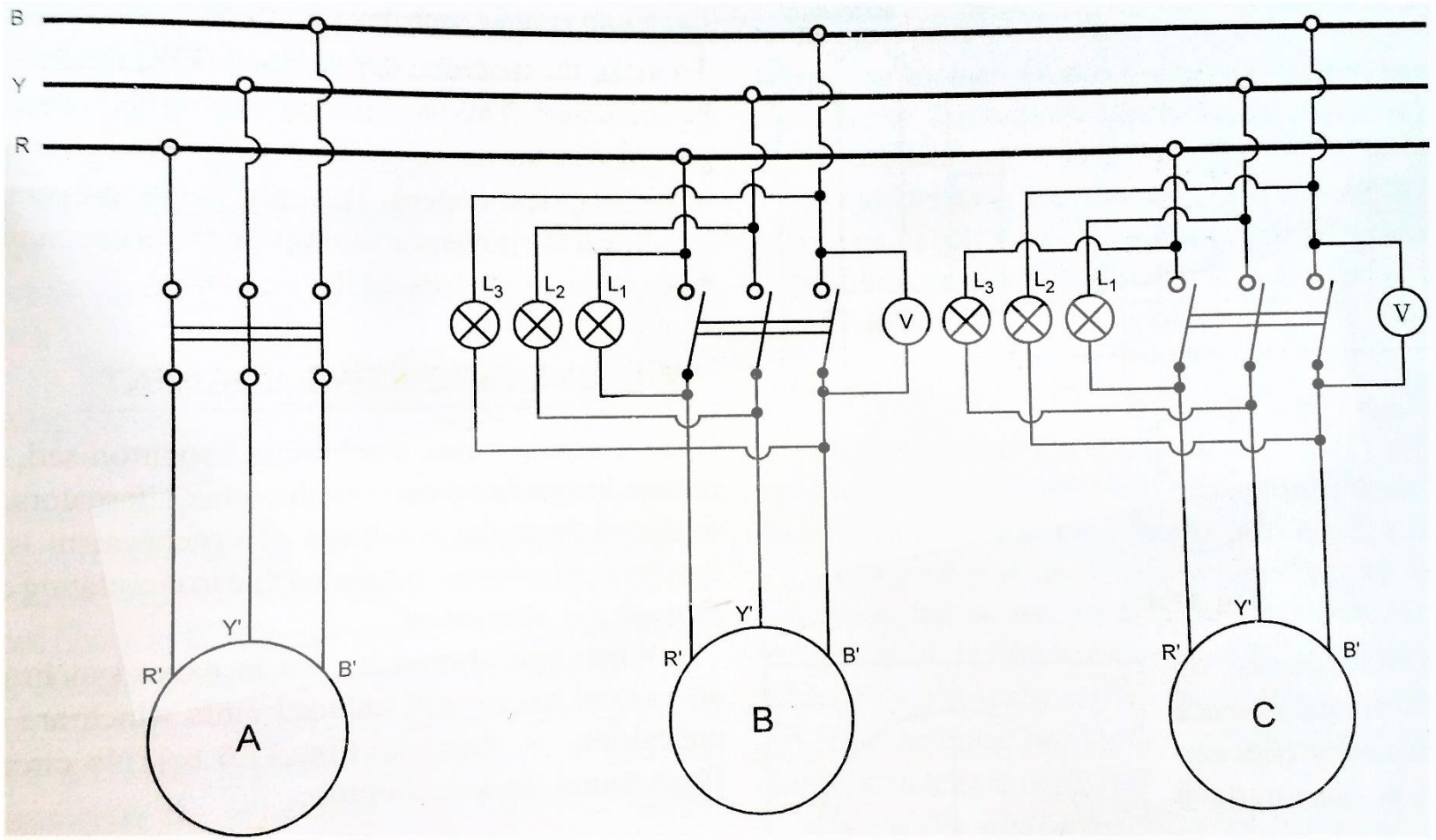
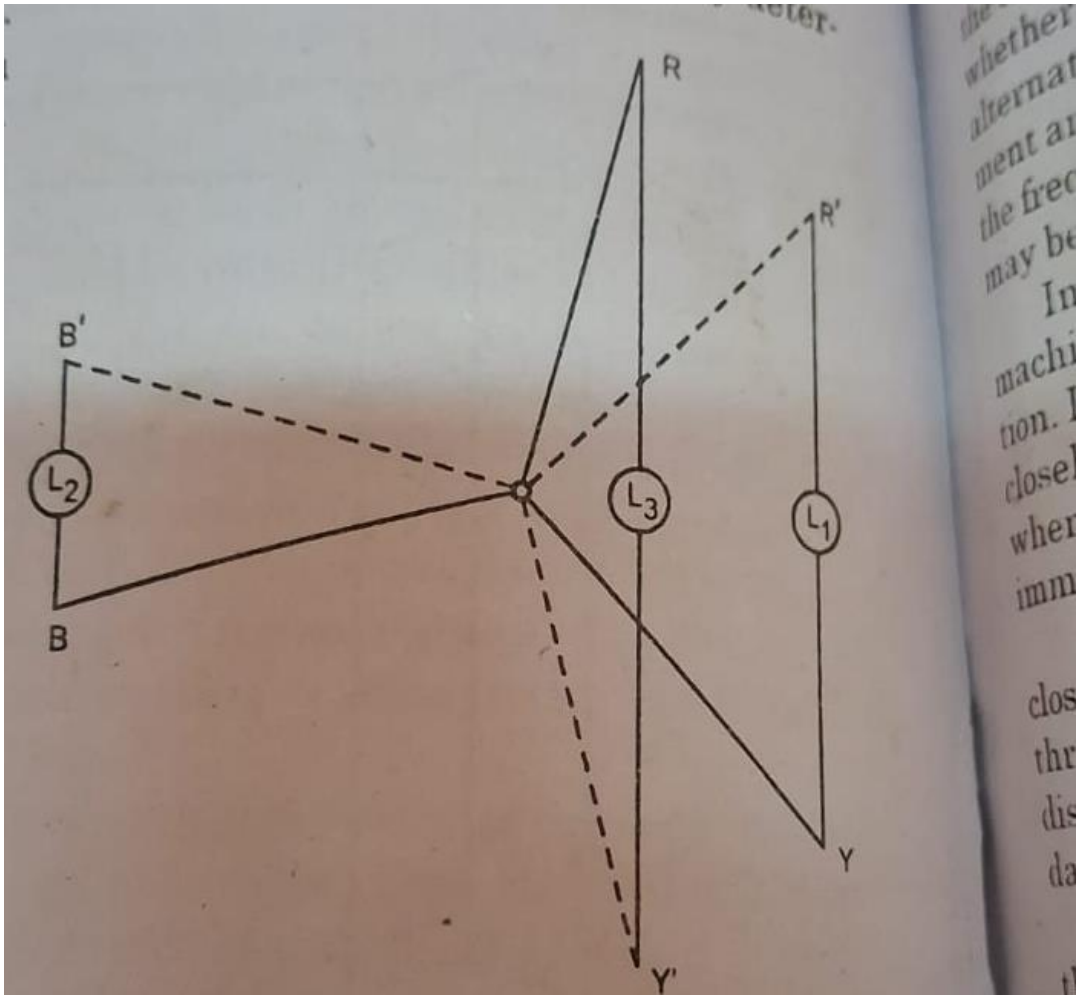


Figure 11 : Synchronising of 3-Phase Alternators

Two Bright and One Dark Lamp Meth

- When incoming alternator is in synchronism with running machine, lamps L_1 and L_3 are bright and L_2 is dark.
- Since near the point of synchronism, the brightness of one lamp increases and of another decreases. The instant at which the incoming machine is in synchronism with the bus-bars can be accurately determined and the paralleling switch will be closed at this instant.
- If the incoming machine is too fast → voltage across L_3 decreases, L_1 increases and L_2 decreases.
- If the incoming machine is too slow → voltage across L_3 increases, L_1 decreases and L_2 increases.
- Hence when the three lamps are placed in a ring, a light wave travelling in counter-clockwise direction indicates that the incoming machine is slow and light wave travelling in clockwise direction indicates that the incoming machine is fast.
- Synchronising switch is closed when changes in light are very slow and at the instant the lamp L is dark.


R
R'
B'
L2
L3
L1
B
Y
Y'

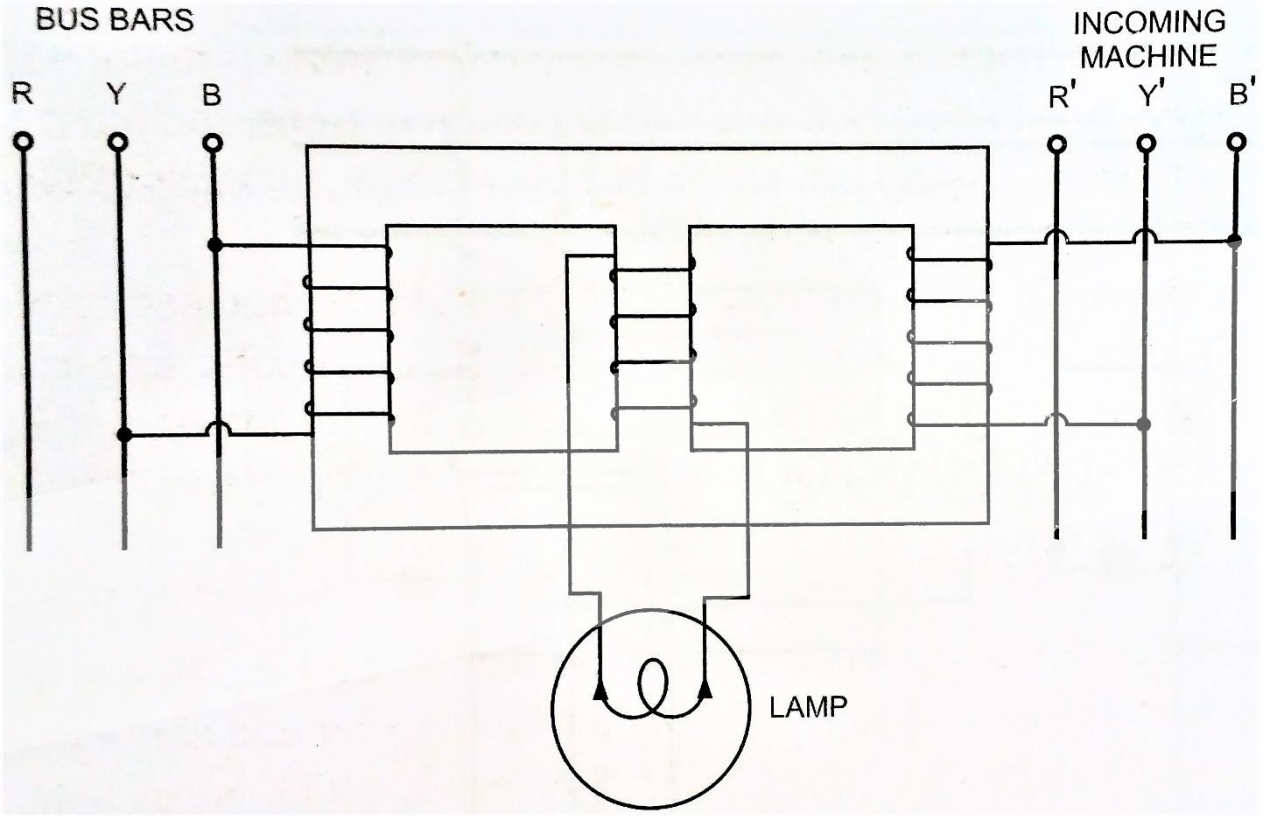


Figure 12 : Three Limbed Transformer and Lamp

Synchroscope

- Synchroscope → Instrument indicating the difference of phase and frequency between two voltages.
- Its a split-phase motor
- Torque is developed if the frequencies of the two voltages differ.
- Voltages from corresponding phases of incoming and running alternators are applie to synchroscope.
- When frequencies are equal, no torque is exerted on the pointer and the pointer stops at the middle of the dial.
- When pointerstops at vertical position, the frequencies are equal, the voltages are in phase and the paralleling switch can be closed.

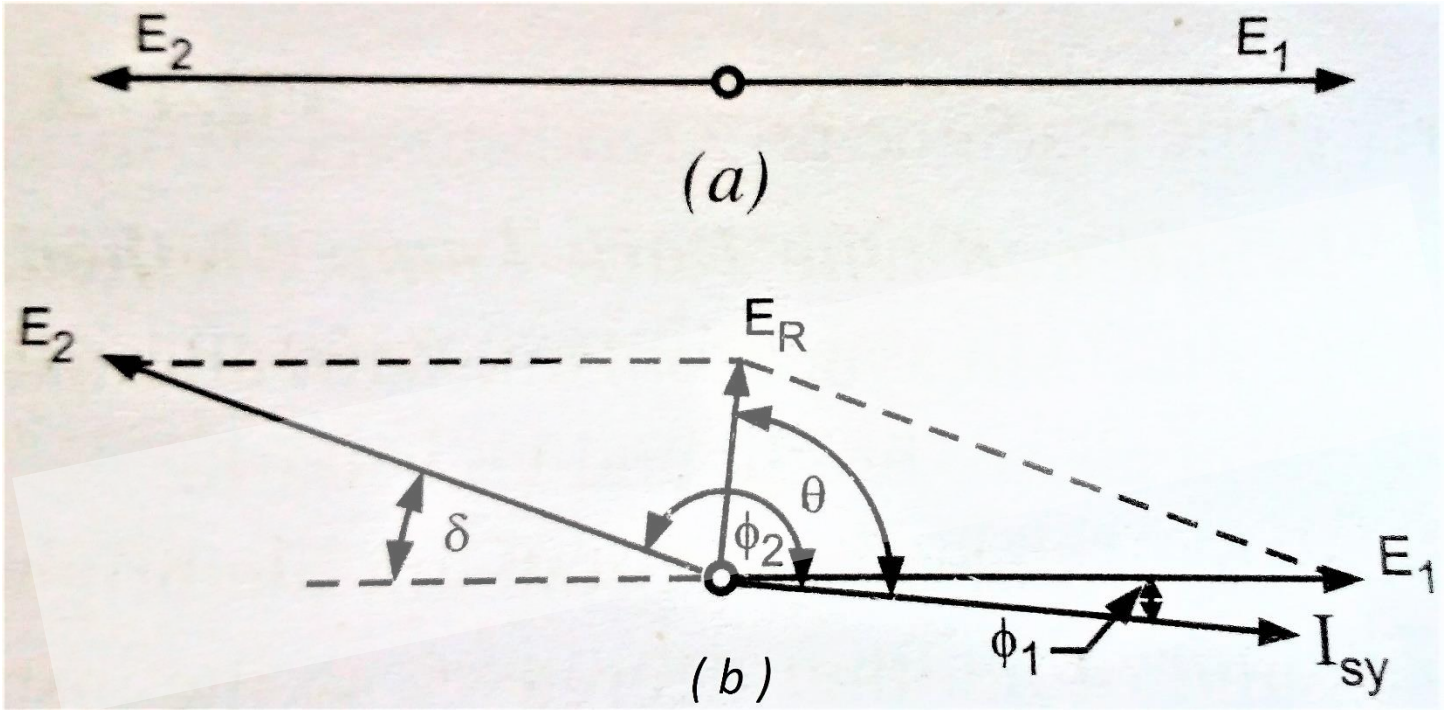


Figure 15 : Voltage Phasors

Synchronising Current

- When two alternators are in synchronism → have equal emfs in exact phase opposition → no circulating current (Fig. (a))
- When induced emfs are equal in magnitude but not in exact phase opposition → resultant emf around the local circuit /= zero → causes flow of current → **Synchronising current(I_{sy})**

  → Let alternator-2 tends to retard,
- E_2 falls back by a phase angle δ electrical degrees
- $|E_1| = |E_2| = E$
  but phase difference = 180 - δ

→ Resultant emf(E_R)

$$E_R = 2E\cos\frac{180 - \delta}{2}$$

$$= 2E\cos 90 - \frac{\delta}{2}$$

$$= 2E\sin\frac{\delta}{2}$$

$$= 2E \times \frac{\delta}{2}$$

$$= E\delta$$

→ Synchronising current(I_{sy})

$$\frac{E_R}{Z_{cs}} = \frac{E\delta}{Z_{cs}}$$

Synchronising Current

- I_{sy} lags behind E_R by an angle θ
- If $R_{ce} <<< X_{cs}$,

$$I_{sy} = \frac{E_R}{X_{cs}}$$

$\Rightarrow$ I_{sy} lags E_R by 90° $\Rightarrow$ almost in phase with E_1

- I_{sy} → generating current w.r.t machine 1 and motoring current w.r.t machine 2
- I_{sy} sets up synchronising torque(T_{sy}) which tends to accelerate machine no.2 and decelerate machine no.1
- Any departure from synchronism results in development of synchronising torque which tends to keep the machines in synchronism

Synchronising Power

- Power supplied by Machine no.1 = $E_1 I_{sy} Cos\Phi_1$
- Power received by Machine no.2 = $E_2 I_{sy} Cos\ (180 - \Phi_2)$
- Power supplied by Machine no.1 = Power received by Machine no.2 + copper losses
- Power supplied by Machine no.1 is called as **Synchronising power(P_{sy})**

$$P_{sy} = E_1 I_{sy} Cos\Phi_1$$

$$= E_1 I_{sy} \not{\delta}$$

$$= E \times \frac{E\delta}{X_{cs}}$$

$$= \frac{\delta E^2}{X_{cs}}$$

- Total synchronising power = $3P_{sy} = \frac{3\delta E^2}{X_{cs}}$

Synchronising Torque

$$3P_{sy} = T_{sy} \times \frac{2\pi N_s}{60}$$

Therefore Synchronising torque,

$$T_{sy} = \frac{3P_{sy} \times 60}{2\pi N_s}$$

where N_s is the synchronous speed in rpm

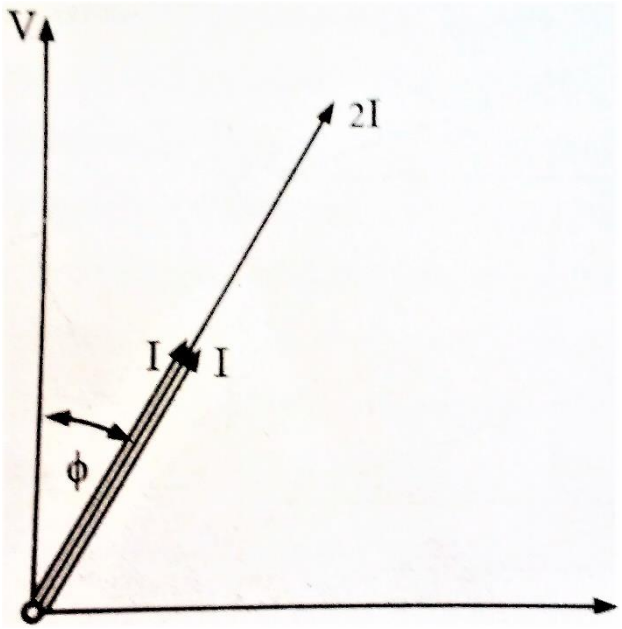


Figure 16 : (a)

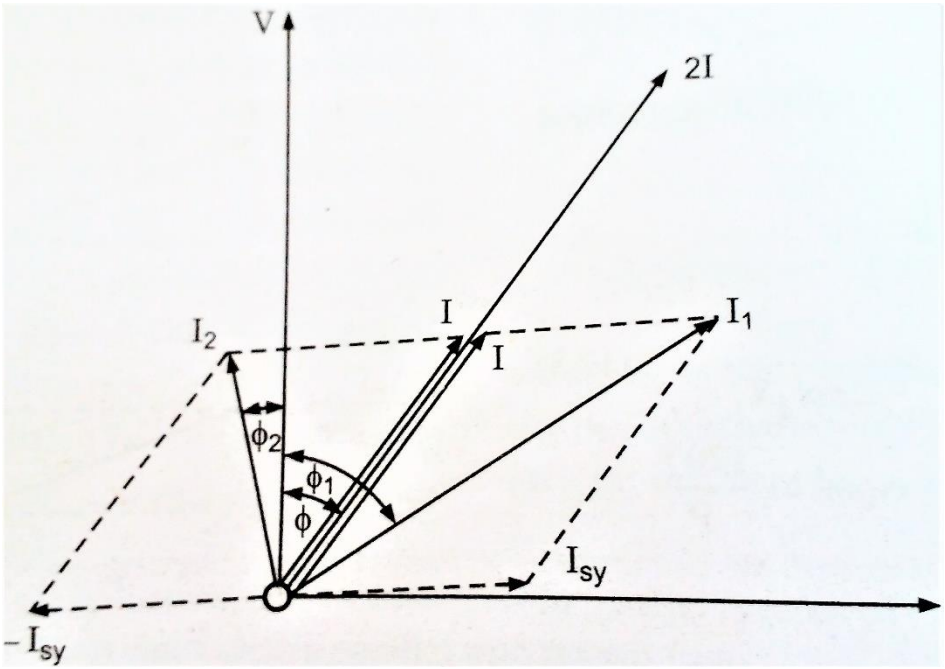


Figure 17 : (b)